

USB-C KIT FOR GAME BOY MICRO



PRODUCT

[HTTPS://SHOP.GILTESA.COM/PRODUCT/GAME-BOY-MICRO-USB-C-KIT](https://shop.giltesa.com/product/game-boy-micro-usb-c-kit)

PLEASE READ THROUGH THESE INSTRUCTIONS ENTIRELY BEFORE ATTEMPTING TO INSTALL

WARNING: IF YOU ARE NOT COMFORTABLE WITH SOLDERING, OR PERFORMING ANY STEP IN THIS GUIDE, DO NOT PERFORM THE INSTALL YOURSELF. FIND SOMEONE WHO IS COMFORTABLE TO DO IT FOR YOU.

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DESCRIPTION

The **Game Boy Micro: USB-C Kit** is a board that allows to replace the original charging connector for a modern and standard USB-C.

If your original connector is too old or damaged and you need a new one, or if you would like to charge your Game Boy Micro with a standard USB-C charger, like the charger of your phone, laptop, you can with this kit.

FEATURES

- Charging your Game Boy Micro with:
 - USB power banks
 - USB-A chargers
 - USB-C chargers
 - USB-C PD chargers (normal speed, not fast)
 - USB-A to USB-C cables
 - USB-C to USB-C cables
- Multiplayer connection support through an adapter.

INCLUDED

- 1 flexible board.

RECOMMENDED / REQUIRED [NOT INCLUDED]

- | | |
|-------------------------------------|---------------------|
| • Tri-wing and Phillips screwdriver | • Desoldering pump |
| • Kapton tape | • Desoldering mesh |
| • Soldering iron | • Isopropyl alcohol |
| • Tin | |
| • Flux | |

BOARD DETAILS

This small flexible board has a total of 10 pads: 4 of them correspond to the USB-C pins/legs, and the others are for power and multiplayer connection.



The pads are, from left to right in the image:

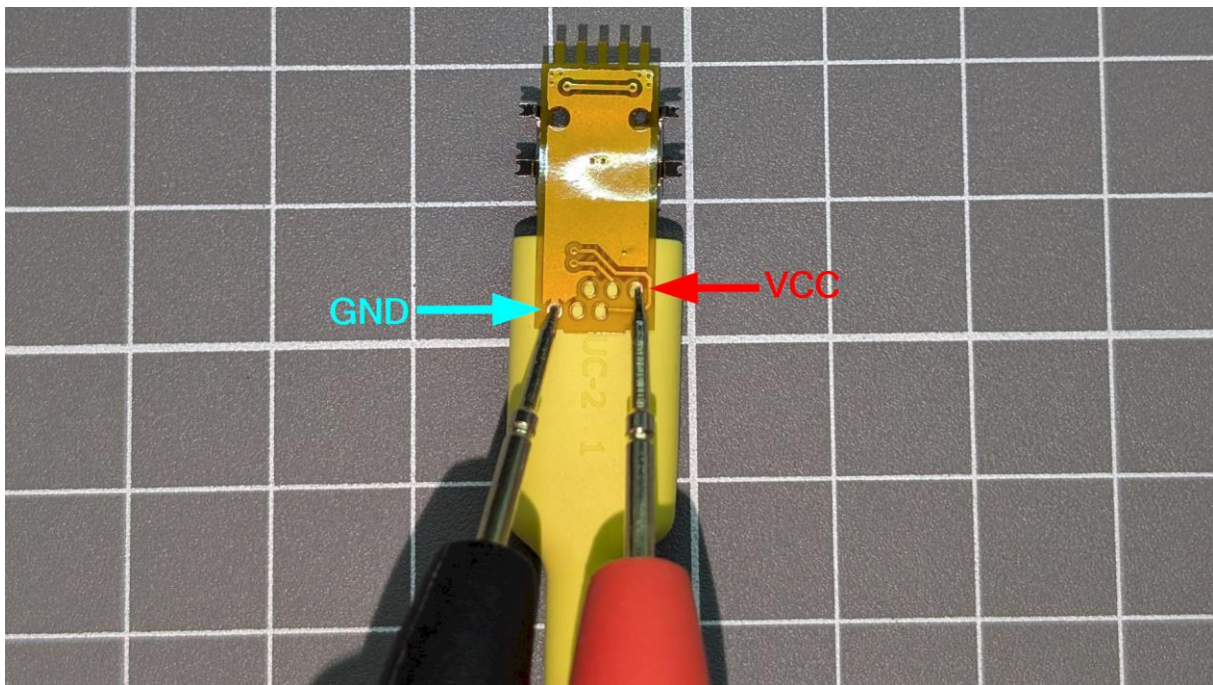
- **VCC** The 5V line from the USB-C.
 - SO Serial output
 - SI Serial input
 - SD Serial data
 - SC Serial clock
 - GND The ground pad.
-
- **USB SHIELD LEGS** They secure the board firmly to the mainboard.

TEST THE BOARD!

Before starting the installation, you should test the board. If it doesn't work contact me [for a replacement](#) (*all boards are fully tested, but they may damage during the shipping, we try to package them as better as possible*), if it works, go ahead with the installation.

Connect the power from your USB charger to the USB-C connector on the board. Then, with a multimeter in voltage measurement mode, **check for a 5V reading. If that's the case, continue with the installation.**

NOTE: Make sure you are making proper contact on the sides of the pads.



Unfortunately, it is not possible to test the data connection until the board is fully installed.

INSTALLATION STEPS

Please, carefully read the following steps for a successful installation.

PRE INSTALLATION STEPS

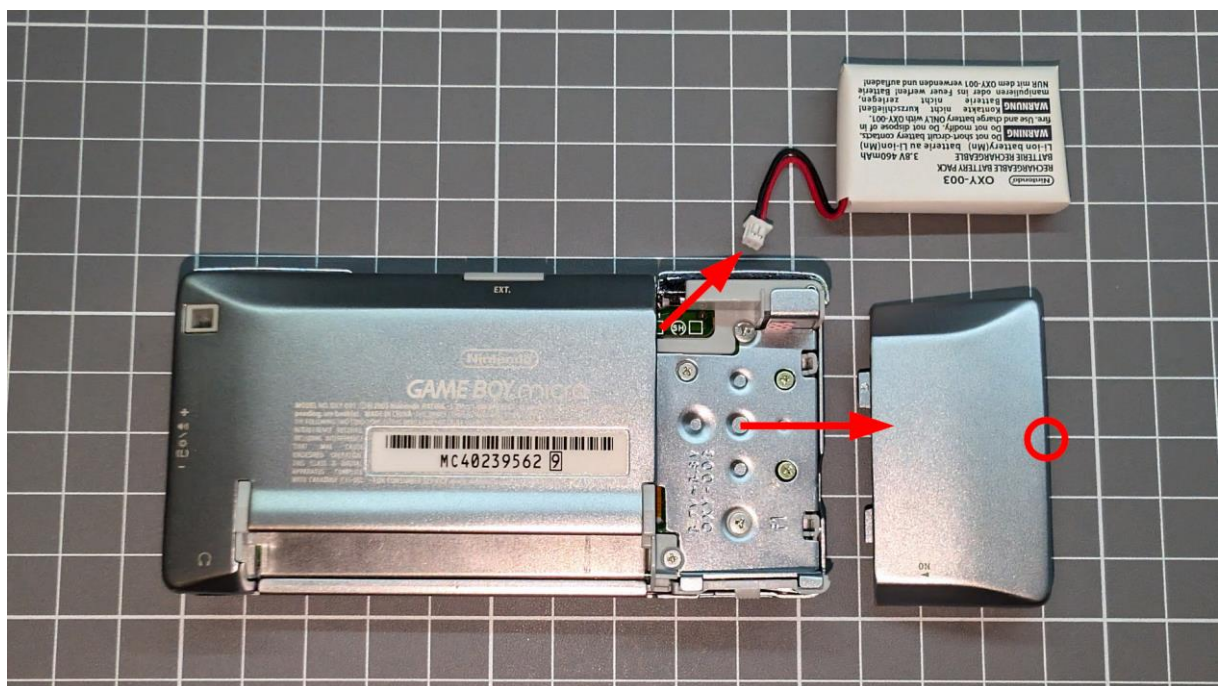
Before the installation, your Game Boy Micro may need some extra steps to have it ready for the kit.

1. DISASSEMBLY THE GAME BOY MICRO

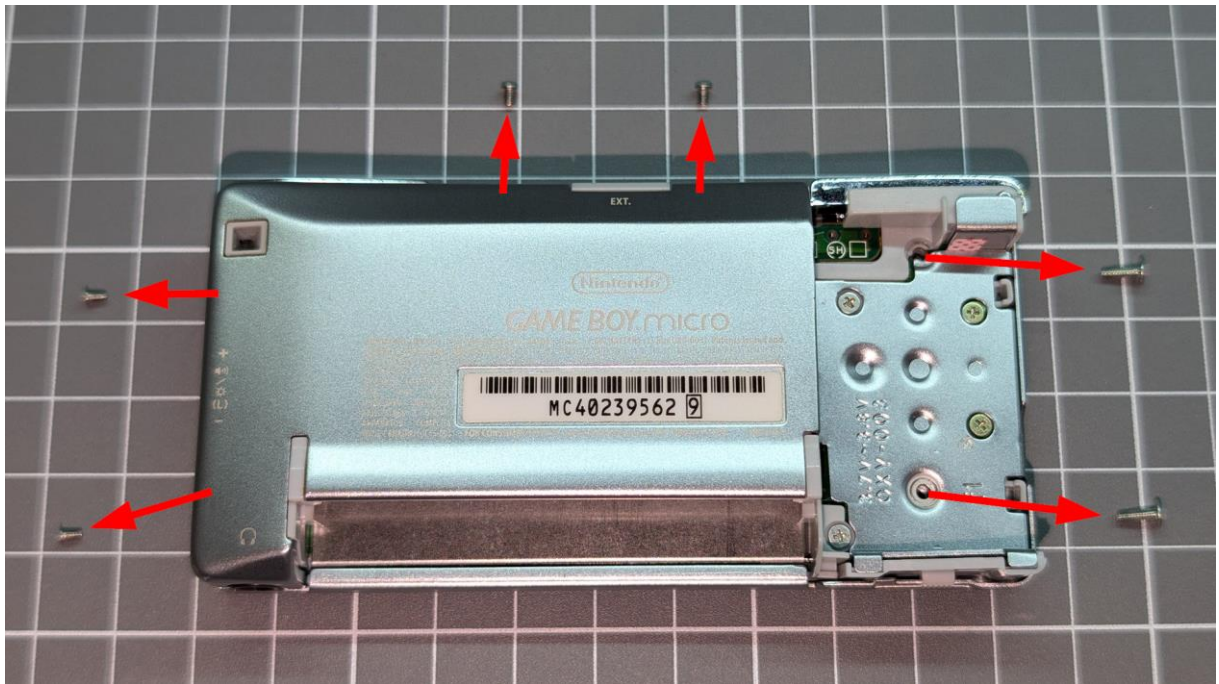
If the steps in this guide are insufficient and you can't proceed, you can refer to the [iFixit guide](#) for opening the console.

This is the most delicate console ever made by Nintendo. When you open and handle it, do so with extreme care. The GBM mainboard consists of a rigid-flex main board and two secondary boards. These three boards are actually just one piece, so do not try to separate them or twist the connections between them.

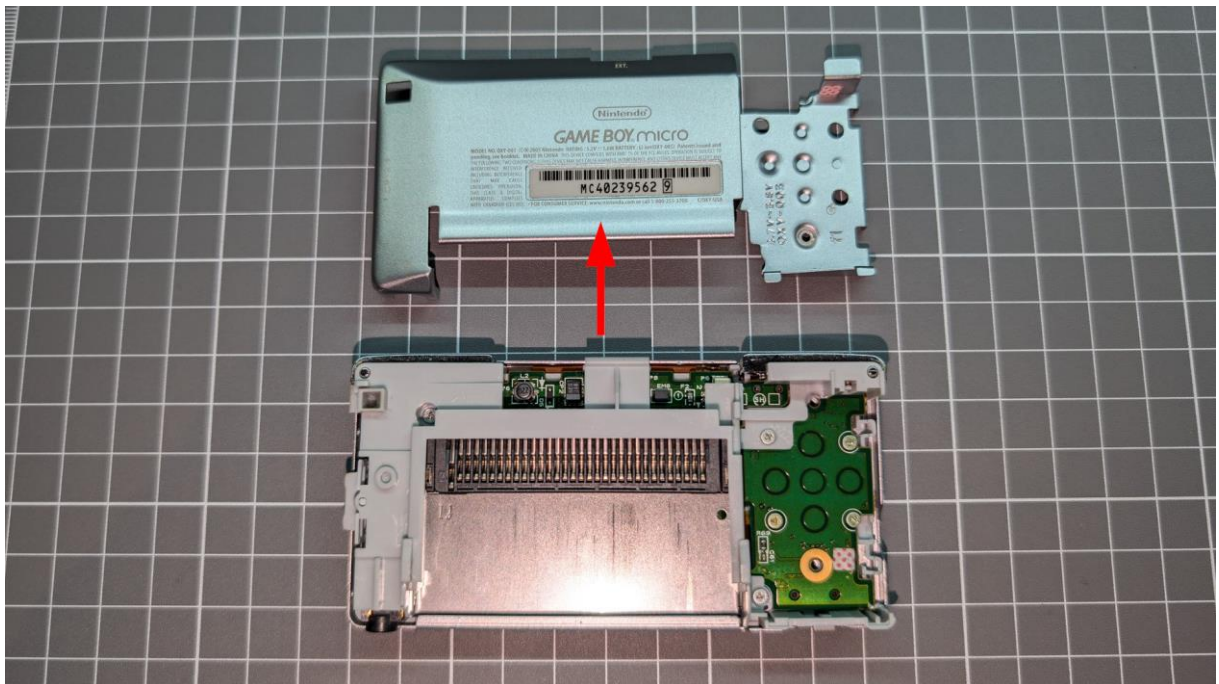
First, remove the side screw holding the battery cover, and once done, disconnect the battery.



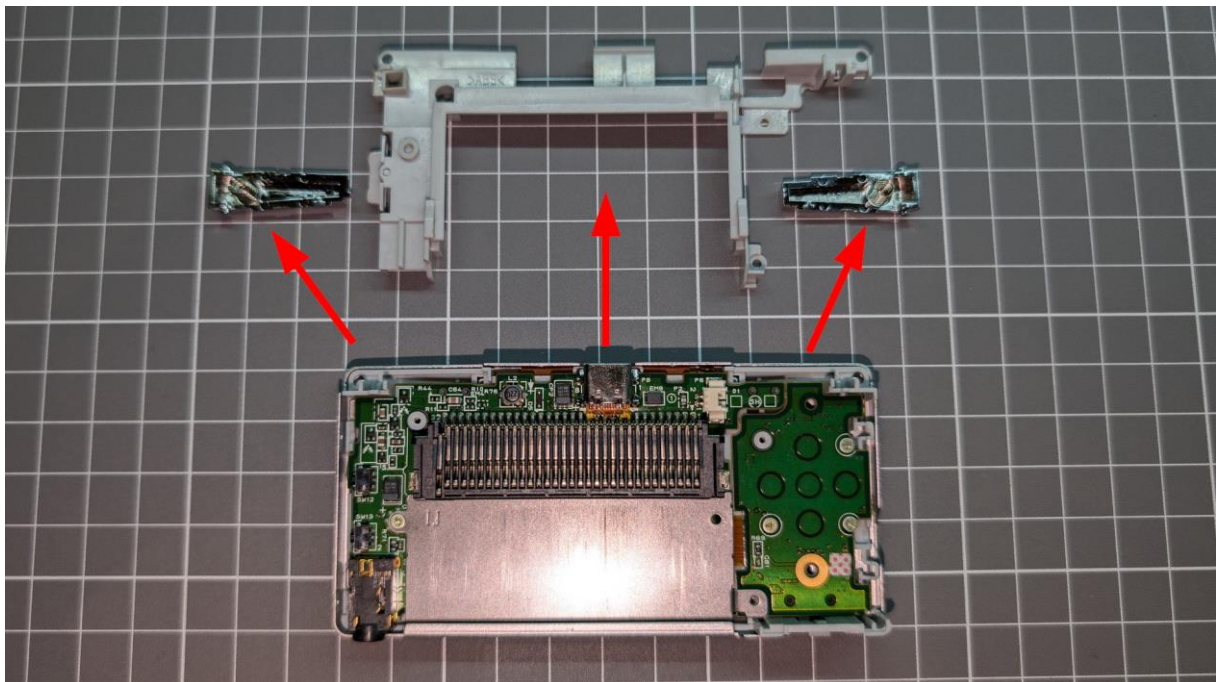
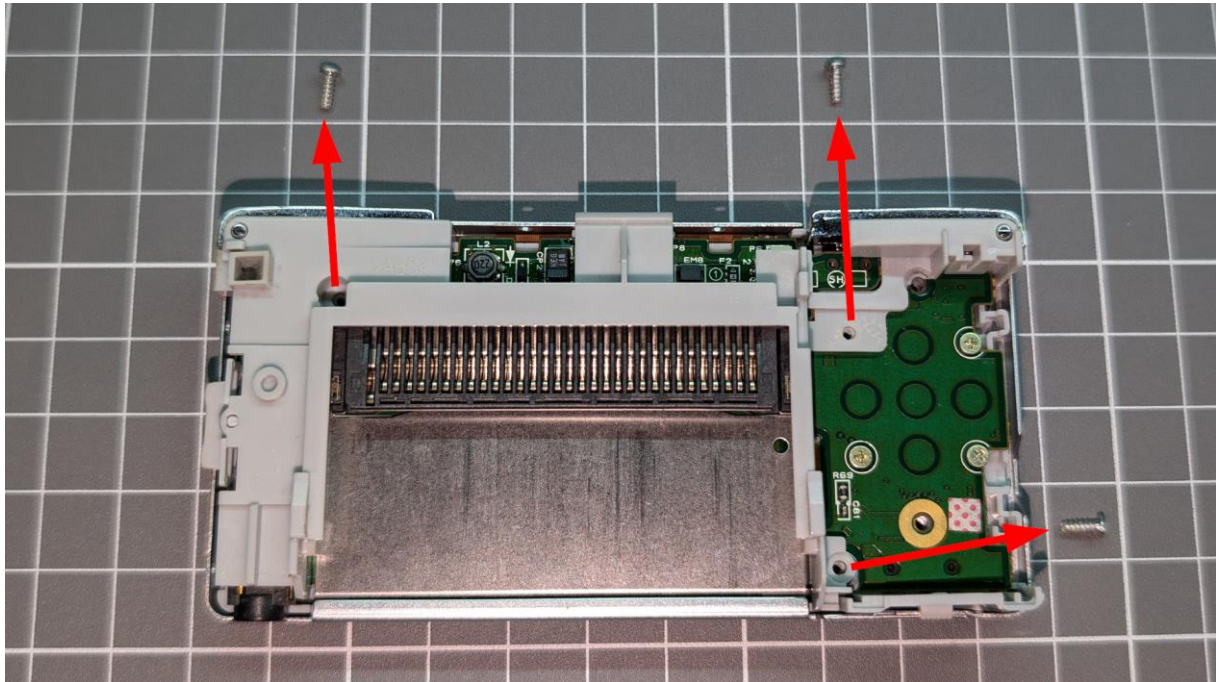
Continue removing the remaining side screws and those in the battery compartment (only the marked ones).



Now you can set aside the back cover of the console.



Remove the three screws holding the plastic frame, as well as the L and R buttons.



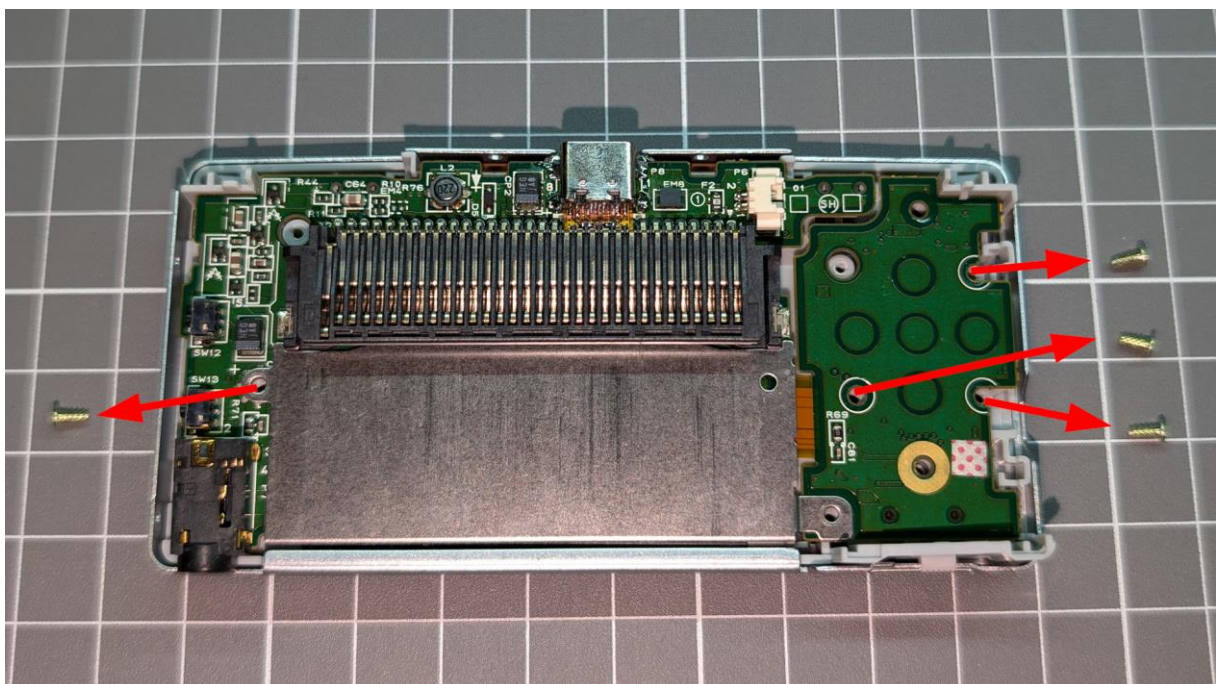
In some images, you will see that the charging connector had already been replaced with the kit's connector; ignore that detail :)

Remove the front faceplate of the console. When placing the console down, do so on a flat surface to avoid scratching the screen.

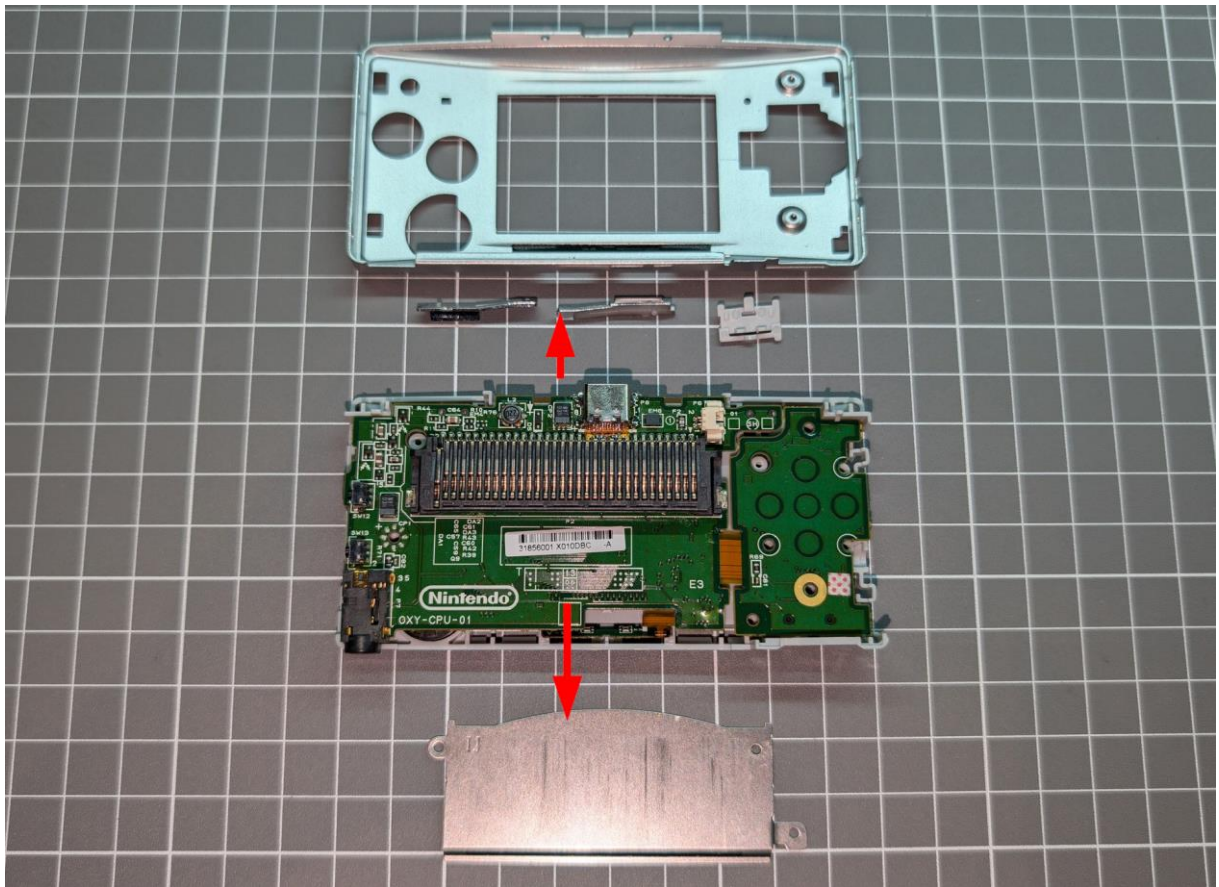
If you don't know how to remove it, take a look at [this video](#).



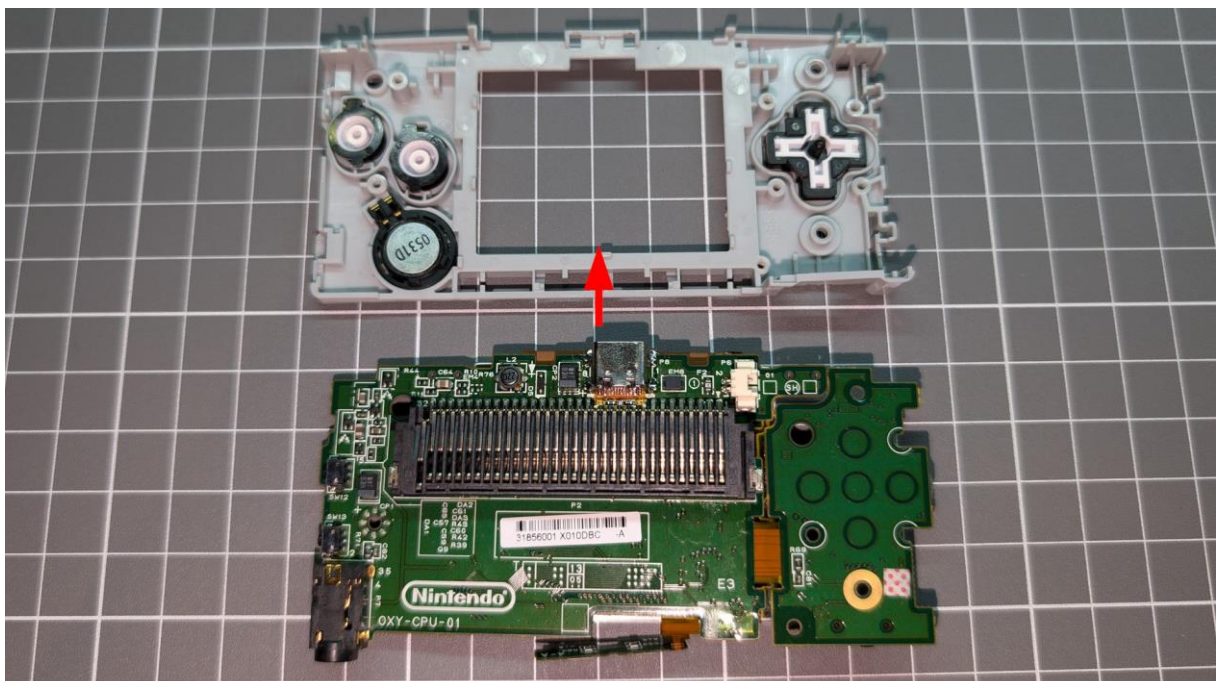
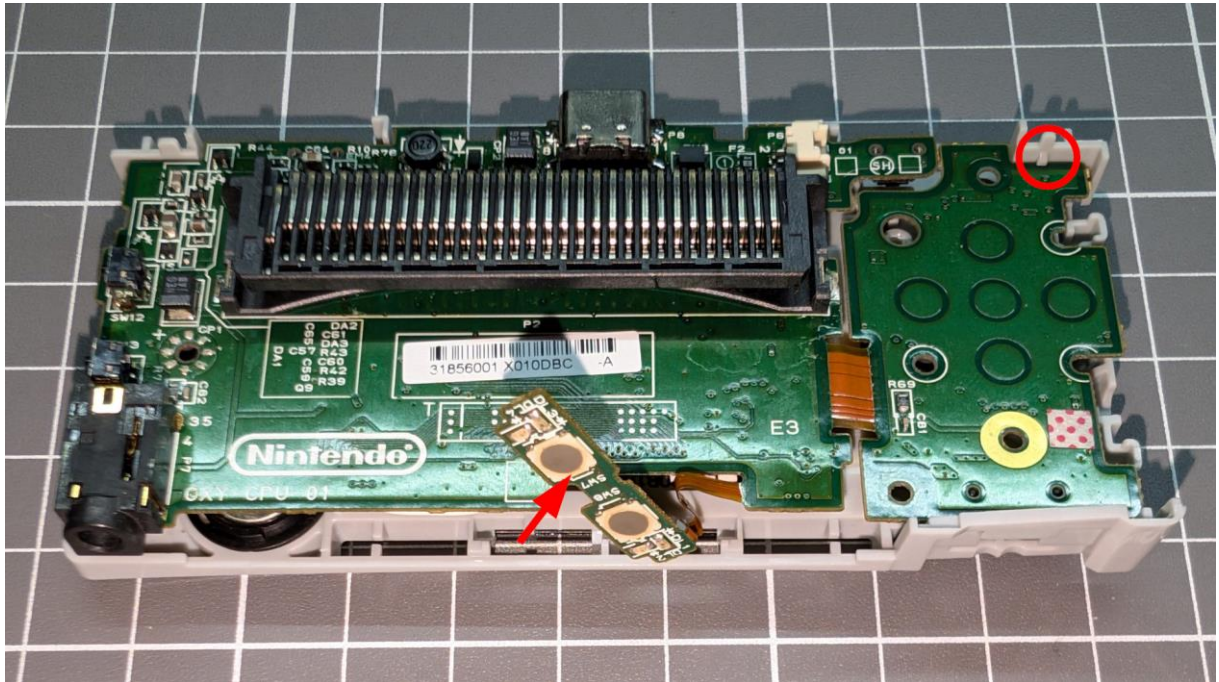
Remove the last four screws holding the mainboard to the bottom plastic frame.



Now you can separate the mainboard from the front casing. To do this, use one hand to pinch the top of the mainboard and the screen at the same time. With your other hand, slowly move the casing in the opposite direction.



Finally, remove the mainboard from the plastic frame that holds its three sections. To do this, first unclip the board from the Select and Start buttons. Then, move on to the board of the D-pad, just enough to surpass the plastic stop that holds it in place. Finally, remove the mainboard while the other two sub-boards separate without resistance.

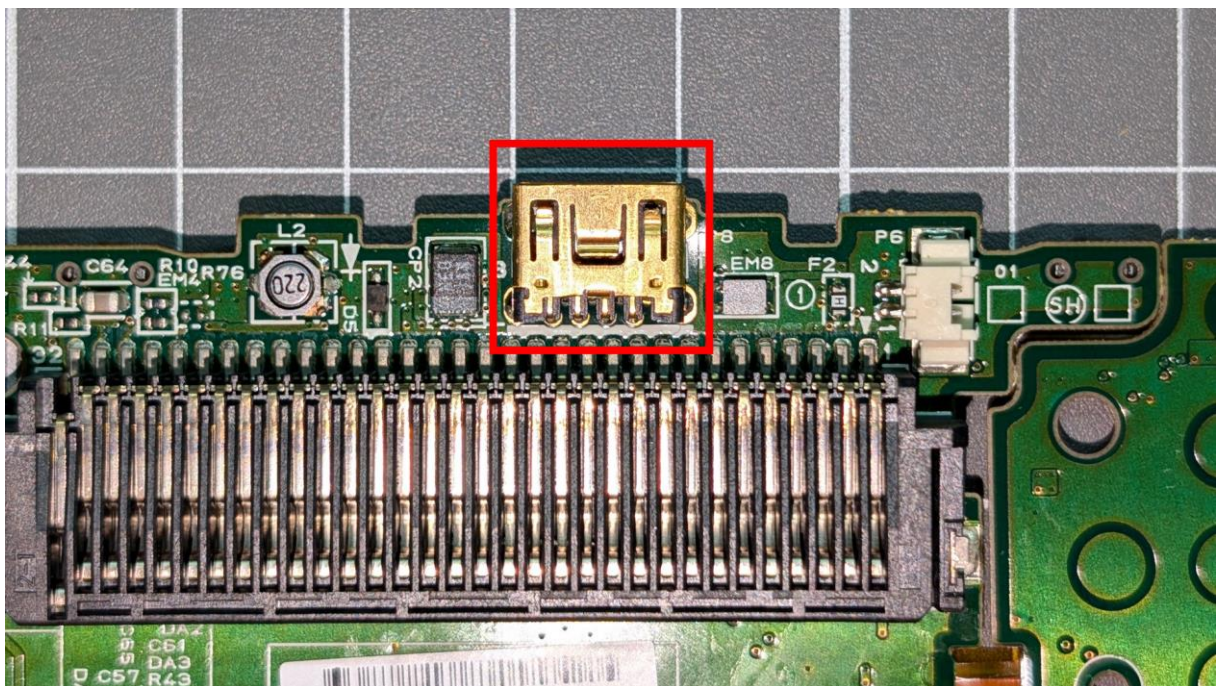
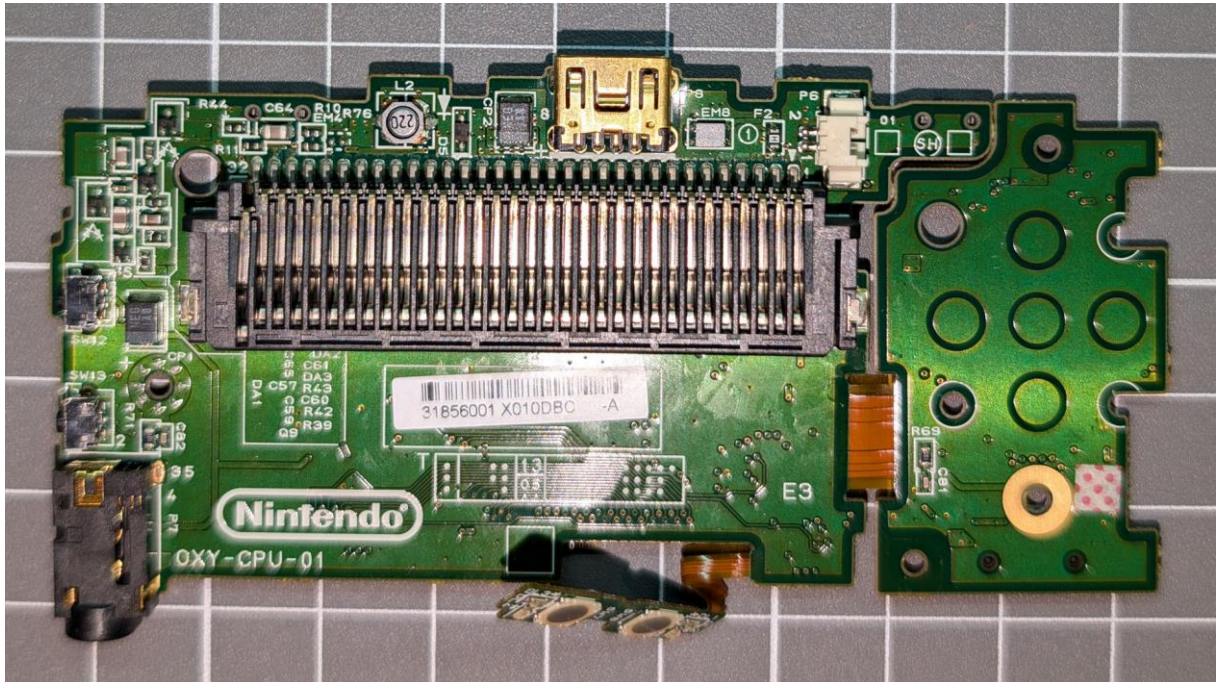


Don't forget to disconnect the screen as well, it is connected to the mainboard with two cables.

INSTALLATION STEPS

1. REMOVE UNNECESSARY COMPONENTS

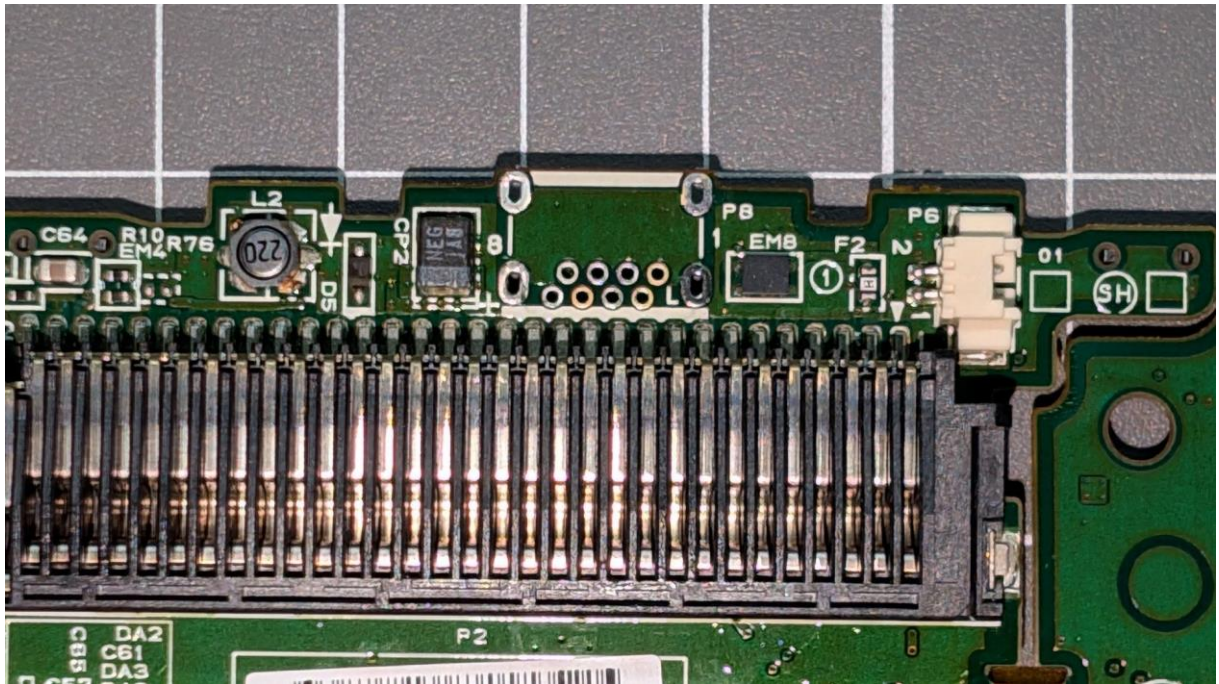
This is the main circuit of the console, and this is the connector that needs to be desoldered to be replaced with the USB-C board.



Before even turning on the soldering iron or the hot air station, thoroughly protect all the nearby areas with kapton tape. Don't skimp on the kapton tape.

Now, remove the connector. This can be challenging because the twelve legs of the connector's housing are firmly attached to the mainboard.

Apply flux and desolder each leg one by one. You can also add higher-quality solder to mix with the existing solder, which will help you remove it all more easily.



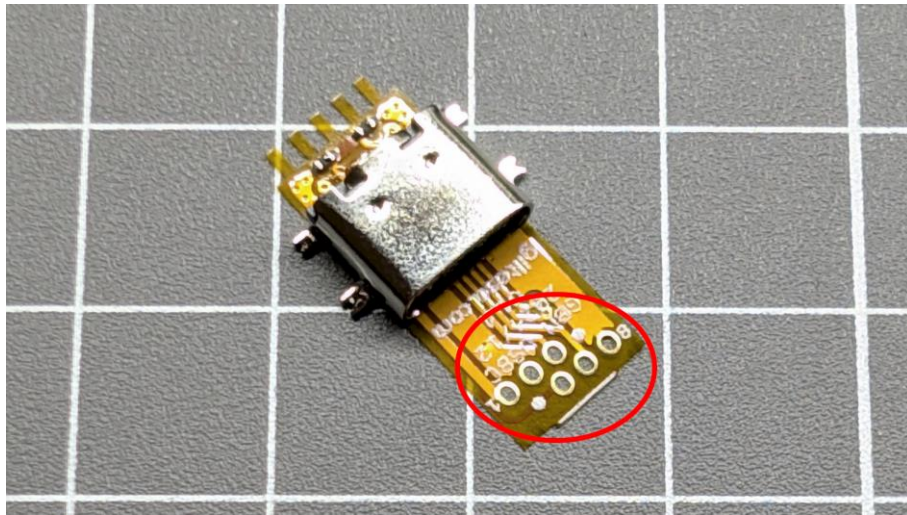
Remove the solder from the pins, ensuring they are not bridged together. It's not necessary to remove the solder from the four holes where the connector was soldered but try to ensure that the solder/pads are completely flat.

2. INSTALLATION OF THE USB-C BOARD

It's time to install the USB-C board.

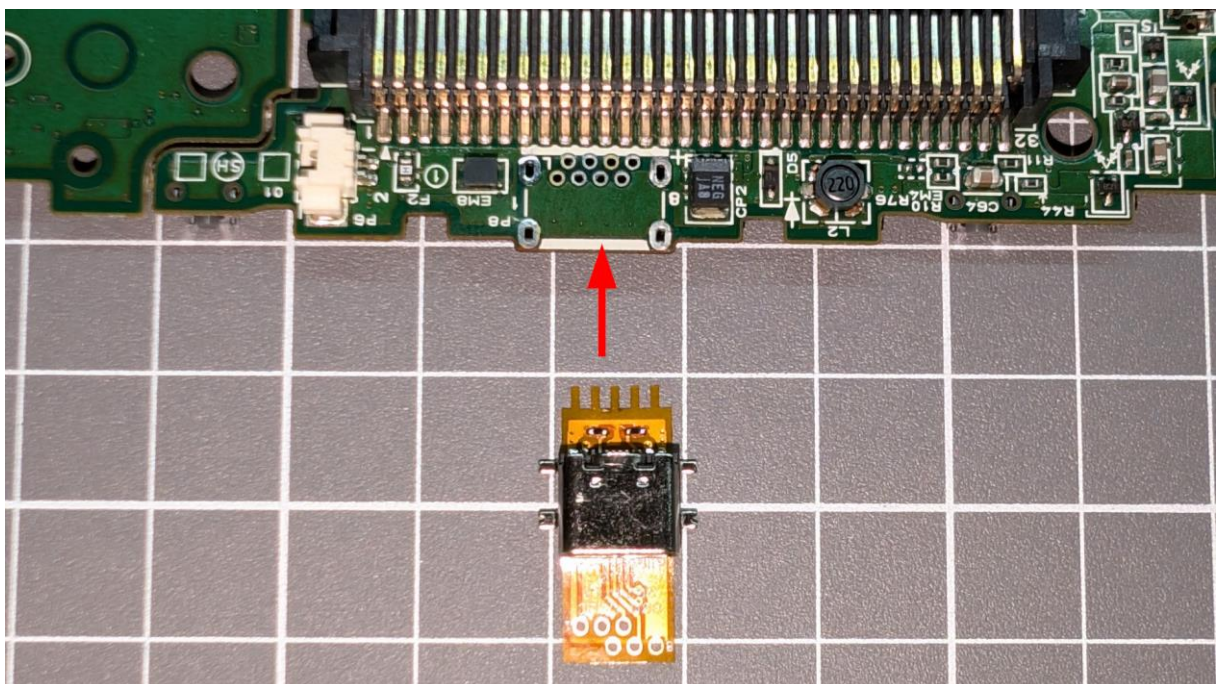
PREPARE THE BOARD FOR THE INSTALLATION

First, press the 6 flexible circuit pads; this will make it easier to solder the board later.

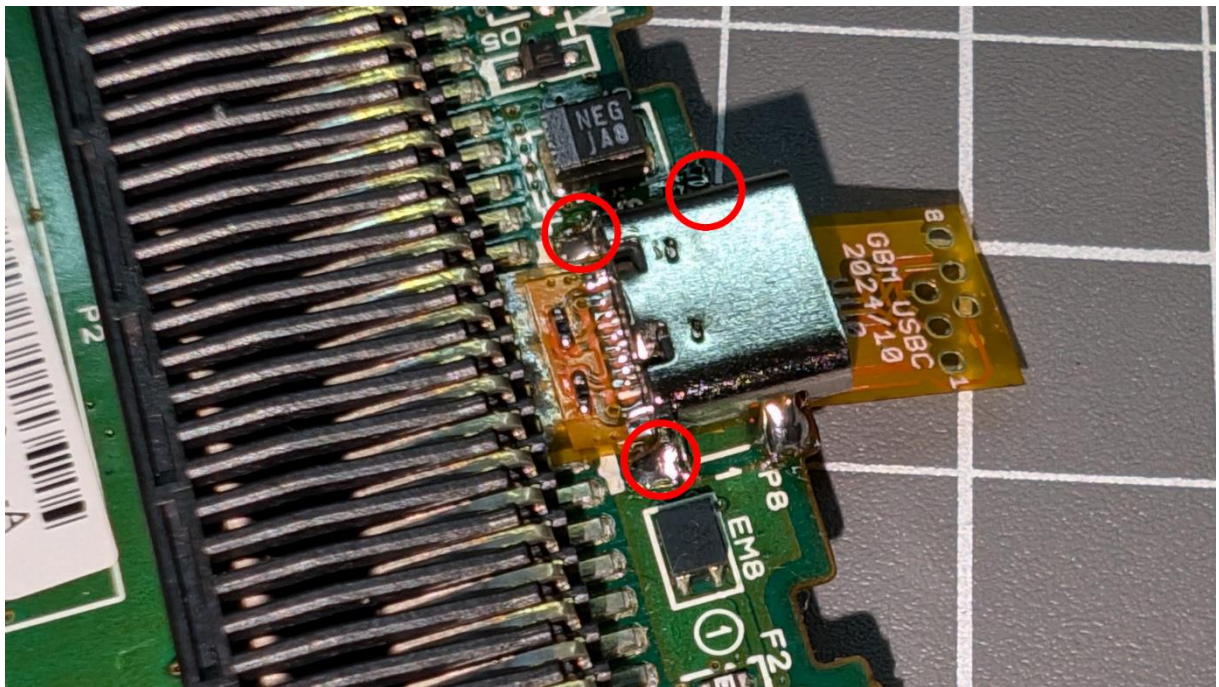
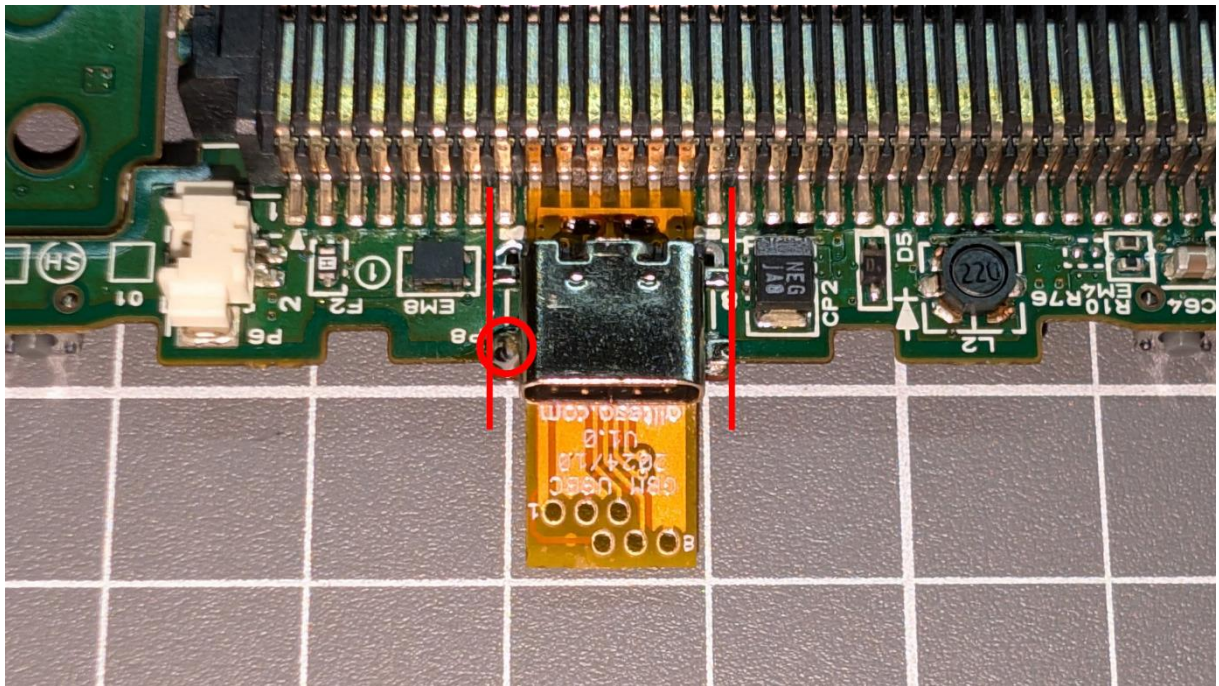


BOARD INSTALLATION

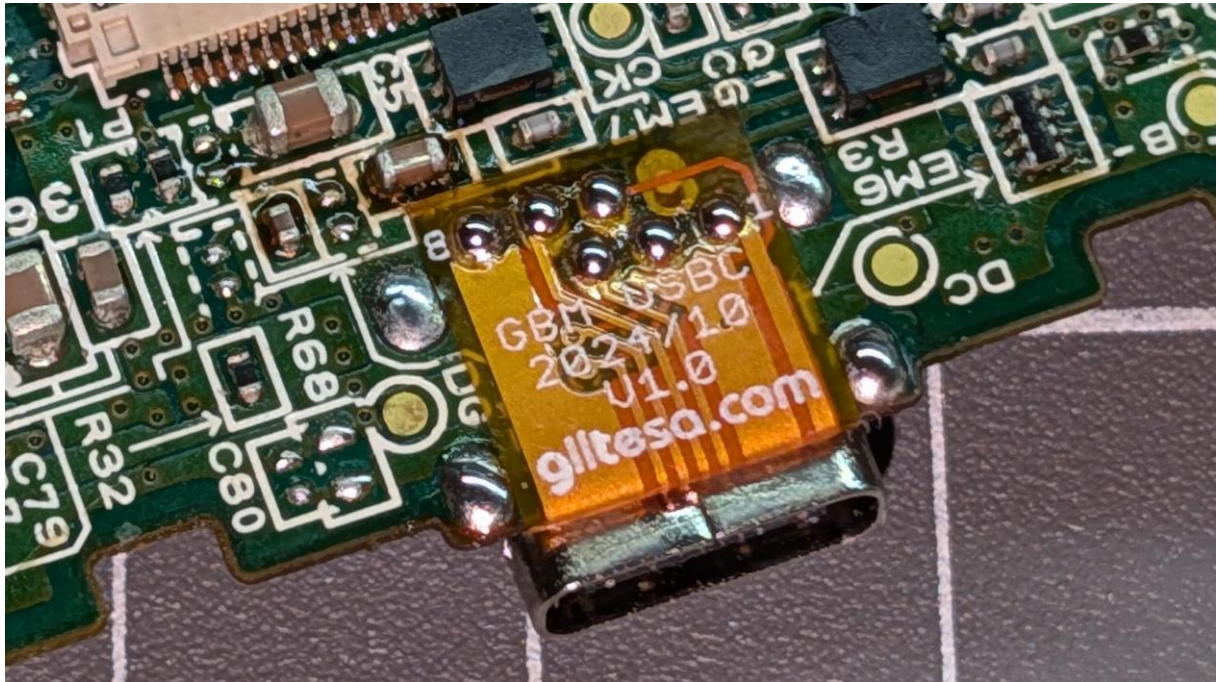
Center the board in its position. You will see that the top "comb" of the flexible board aligns with the slots of the game reader pins. This will help you position it as accurately as possible. Even so, ensure that the connector is perfectly centered.



Once the connector is centered, solder one of its pins/legs. Check again to ensure it remains centered, then solder the rest of the pins/legs to the mainboard.

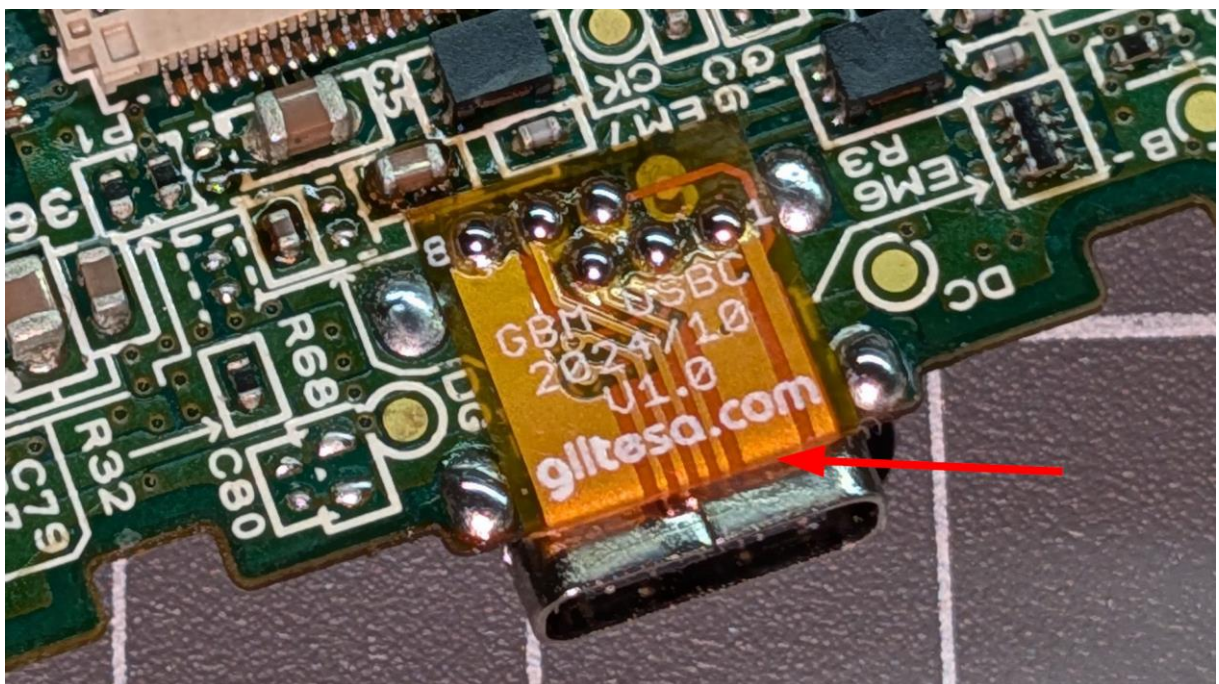


After completing the soldering, clean the boards with isopropyl alcohol.



It is recommended to use a black permanent marker to color the outer edge of the flex PCB. This will make it invisible once the console is reassembled.

We forgot to do it in this installation, but we did it in others, and it noticeably improves the final result.



3. FINISHING THE INSTALLATION

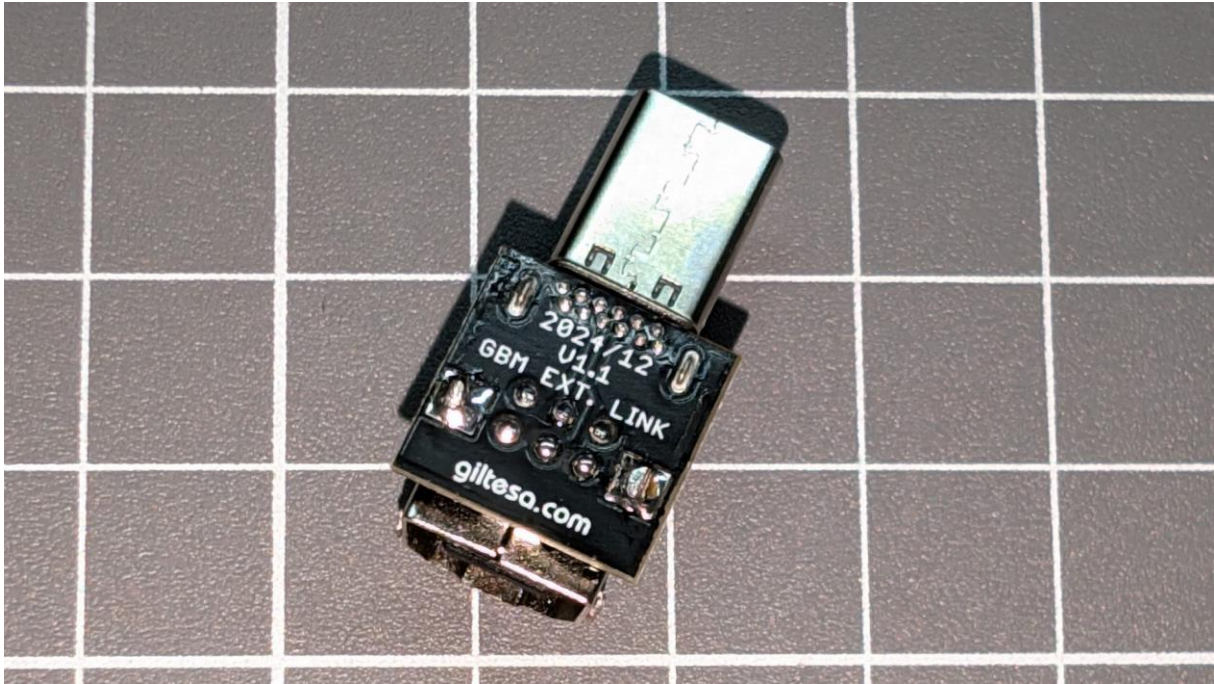
Follow the steps backward, placing each thing in its place, and the installation will be complete. Enjoy it!



MULTIPLAYER ADAPTER

Since the original GBM connector provides two functionalities: charging the battery and transferring data to another GBM for multiplayer gaming. It is now necessary to use a USB-C to Link Cable adapter.

The adapter comes fully assembled. Inside, there is a circuit with a USB-C male connector on one end and a Game Boy Advance Link Cable connector on the other.



When you connect it to the GBM, the text must face downward; otherwise, the multiplayer connection will not work.



FREQUENTLY ASKED QUESTIONS - FAQ

WHAT CHARGER CAN BE USED?

You can use any standard charger for mobile phones, computers, etc., with 5V 1A. It doesn't need to be a Power Delivery charger since this feature is not used. Of course, if you want to use a Power Delivery charger, there's no problem or risk.

Technical data for curious minds:

Power Delivery chargers can supply a wide range of voltages: 5V, 9V, 12V, 15V, and 20V. However, for this to happen, the device must communicate with the charger to explicitly request the desired voltage. Without this communication, the charger will never supply more than 5V. That's one of the advantages of USB-C, as it can be used with both old and modern devices.

MY KIT DOES NOT INCLUDE A PLASTIC CAP TO COVER THE HOLE.

The photos may make it look like there's a plastic cap installed, but that's not the case. It's actually the original plastic of the console. Depending on your console's color, the plastic will vary accordingly.

